



**PROGRAMMING FUNDAMENTALS (CCP6114)**

**TRIMESTER OCTOBER (T2430)**

**Assignment Programming (TC1L\_TT1L\_Group1)**

**User Documentation for Table Management Application**

| ID         | Name                 | Email                              | Phone         |
|------------|----------------------|------------------------------------|---------------|
| 242UC241BQ | ALVIN LO JIAN WEN    | ALVIN.LO.JIAN@student.mmu.edu.my   | 011-2741 6523 |
| 242UC244M6 | YAP HUI CHI          | YAP.HUI.CHI@student.mmu.edu.my     | 016-3225726   |
| 242UC2426R | BIANCA LAU YING XUAN | BIANCA.LAU.YING@student.mmu.edu.my | 010-2752246   |
| 242UC244KQ | NG BAI KEONG         | NG.BAI.KEONG@student.mmu.edu.my    | 018-6605595   |

## **Table of Contents**

1. INTRODUCTION
2. SYSTEM REQUIREMENTS
3. INSTALLATION
4. GETTING STARTED
  - 4.1 LAUNCHING THE APPLICATION
  - 4.2 MAIN MENU OVERVIEW
5. FEATURES AND USAGE
  - 5.1 CREATE A NEW FILE
  - 5.2 OPEN AN EXISTING FILE
  - 5.3 EXIT THE APPLICATION
  - 5.4 SECONDARY MENU OPERATION
    - 5.4.1 DELETE A SPECIFIC ROW
    - 5.4.2 UPDATE A SPECIFIC FIELD
    - 5.4.3 COUNT THE NUMBER OF ROWS
    - 5.4.4 INSERT A NEW ROW
    - 5.4.5 BACK TO MAIN MENU
    - 5.4.6 EXIT THE APPLICATION
6. TROUBLESHOOTING
7. FREQUENTLY ASKED QUESTIONS (F&Q)
8. CONTACT SUPPORT

## 1.0 INTRODUCTION

Welcome to the **Light Mariadb Interpreter**, a command-line tool designed to help you create, manage, and manipulate table data stored in text files. Whether you are maintaining simple datasets or managing more complex information, this application provides an intuitive interface to handle your data efficiently.

## 2.0 SYSTEM REQUIREMENTS

- **Operating System:** Windows, Linux, or macOS
- **Compiler:** A standard C++ compiler
- **Development Tools (Optional):** Any text editor or IDE for C++ development
- **Dependencies:** Standard C++ libraries

## 3.0 INSTALLATION

1. **Download/Save the Source Code:**  
Save the complete source code into a file
2. **Open a Terminal or Command Prompt:**  
Navigate to the directory containing your source file.
3. **Compile the Application**
4. **Verify the Executable:**  
Ensure that the executable file has been created in your working directory.

## 4.0 GETTING STARTED

### 4.1 LAUNCHING THE APPLICATION

- **Windows:** Double-click the TableManagement.exe file or run it via the command prompt.

Once launched, the application will display the main menu with a list of options for file creation, file opening, and existing the program.

### 4.2 MAIN MENU OVERVIEW

Upon startup, the main menu presents three primary options:

1. **Create a file (.txt):**  
Start a new table by creating a text file and defining its structure.
2. **Open existing file (.mdb):**  
Load data from an existing CSV-formatted .mdb file and convert it into a .txt file.
3. **Exit:**  
Terminate the application.

```
1. Create a file
2. Open existing file
3. Exit
Enter your choice:
```

Choose the desired option by entering the corresponding number.

## 5.0 FEATURES AND USAGE

### 5.1 CREATE A NEW FILE

When you choose to create a new file:

- **File Name:** You will be prompted to enter a file name (without any extension). The program creates a new file with a .txt extension.
- **Table Definition:**  
You are asked to provide a table name (e.g., customer) and the number of columns. For each column, enter a short column name (e.g., id, name, city). The program automatically prefixes each short name with the table name (e.g., customer\_id).
- **Initial Data Entry:** Specify the number of rows you want to add initially and then input data for each column in every row.
- **File Writing:** The table structure (including column headers and rows) is written in CSV format to <fileName>.txt.
- **Display:** The table content is displayed on the screen for confirmation.

### 5.2 OPEN AN EXISTING FILE

When opening an existing file:

- **File Name Prompt:** Enter the file name (without extension) for the file you wish to open. The program will attempt to open <fileName>.mdb.
- **Data Loading:** The application reads the CSV-formatted data from the .mdb file, with the first line as column headers and subsequent lines as data rows.

```
customer_id,customer_name,customer_city,customer_state,customer_country,customer_phone,customer_email
1,name1,city1,state1,country1,phone1,email1
2,name2,city2,state2,country2,phone2,email2
3,name3,city3,state3,country3,phone3,email3
4,name4,city4,state4,country4,phone4,email4
```

- **File Conversion:** The loaded data is written to a new <fileName>.txt file.
- **Display:** The table content is then displayed on the console.

### 5.3 EXIT THE APPLICATION

- **Main Menu Exit:** Selecting the "Exit" option (option 3) in the main menu will terminate the program immediately.

## 5.4 SECONDARY MENU OPERATION

After creating or loading a table, the application transitions to a secondary menu that offers additional operations on the table data. The options in this menu allow you to manage and modify the data in real time.

```
Choose an option:
1. Delete a specific row
2. Update a specific field
3. Count the number of rows
4. Insert a new row
5. Back to main menu
6. Exit
Enter your choice:
```

### 5.4.1 DELETE A SPECIFIC ROW

- **Operation:**  
Choose the delete option to remove a row from the table.
- **Steps:**
  1. The current table content is displayed with row numbers.
  2. Enter the row number you wish to delete.
  3. The application verifies the row number and, if valid, deletes the row from memory.
  4. The updated table is saved back to the file and displayed.

### 5.4.2 UPDATE A SPECIFIC FIELD

- **Operation:**  
Update a field in a specific record by using its **id**.
- **Steps:**
  1. The current table content is displayed.
  2. Enter the **id** of the record to update (the program assumes that the id is located in the first column).
  3. If a record with the specified id is found, the record is displayed with numbered columns.
  4. Enter the column number corresponding to the field you wish to update.

5. Input the new value for the chosen field.
6. The application updates the record in memory, writes the changes back to the file, and confirms the update.

#### **5.4.3 COUNT THE NUMBER OF ROWS**

- **Operation:**  
Count and display the total number of data rows (excluding the header).
- **Steps:**
  1. Selecting this option will show the current count of rows in the table.

#### **5.4.4 INSERT A NEW ROW**

- **Operation:**  
Insert a new row into the table.
- **Steps:**
  1. The application displays the current table structure.
  2. Enter data for each field in the new row.
  3. The new row is appended to the table.
  4. The updated table is written back to the file and displayed.

#### **5.4.5 BACK TO MAIN MENU**

- **Operation:**  
Return from the secondary menu to the main menu.
- **Steps:**
  1. Select this option if you wish to go back to the main menu and choose another operation (e.g., create a new file or open an existing file).

#### **5.4.6 EXIT THE APPLICATION**

- **Operation:**  
Terminate the program directly from the secondary menu.
- **Steps:**
  1. Selecting this option will exit the application immediately.

## 6.0 TROUBLESHOOTING

- **File Already Exists:**

- **Issue:** When creating a new file, you may encounter an error if the file already exists.
- **Solution:** Use a unique file name or delete/rename the existing file.

- **Unable to Open File:**

- **Issue:** The program may fail to open a file if it does not exist or the name is entered incorrectly.
- **Solution:** Ensure that the file name (without extension) is correct and that the file exists in the expected directory.

- **Invalid Row or Column Number:**

- **Issue:** Errors may occur if you enter a row number or column number that is out of range.
- **Solution:** Verify the table display to use valid row and column numbers.

- **Input Format Issues:**

- **Issue:** The application uses `cin >>` for input, which does not support spaces within a single field.
- **Solution:** Enter single-word values or modify the input method to support spaces if necessary.

## **7.0 FREQUENTLY ASKED QUESTIONS (F&Q)**

### **Q1: What file extensions does the application use?**

**A:** The application creates new files with a .txt extension and opens existing files with a .mdb extension.

### **Q2: How is the table structure defined?**

**A:** The user defines the table by specifying a table name, the number of columns, and short column names. The program then automatically creates full column names by prefixing each short name with the table name.

### **Q3: How does the update functionality work?**

**A:** The update function locates a record by its **id** (assumed to be in the first column). Once found, you can select which column (by number) you want to update and enter a new value.

### **Q4: What happens if I enter an invalid row or column number?**

**A:** The program performs input validation and will display an error message if an invalid number is entered. You will then be prompted to re-enter a valid number.

### **Q5: Can I return to the main menu after performing an operation?**

**A:** Yes. In the secondary menu, you have the option to go back to the main menu at any time.

## 8.0 CONTACT SUPPORT

If you encounter issues or have further questions about the application, please contact the support team. The team members for this project are:

- **Alvin Lo Jian Wen**  
Email: [ALVIN.LO.JIAN@student.mmu.edu.my](mailto:ALVIN.LO.JIAN@student.mmu.edu.my)  
Phone: 011-2741 6523
- **Yap Hui Chi**  
Email: [YAP.HUI.CHI@student.mmu.edu.my](mailto:YAP.HUI.CHI@student.mmu.edu.my)  
Phone: 016-3225726
- **Bianca Lau Ying Xuan**  
Email: [BIANCA.LAU.YING@student.mmu.edu.my](mailto:BIANCA.LAU.YING@student.mmu.edu.my)  
Phone: 010-2752246
- **Ng Bai Keong**  
Email: [NG.BAI.KEONG@student.mmu.edu.my](mailto:NG.BAI.KEONG@student.mmu.edu.my)  
Phone: 018-6605595

For additional assistance or to report bugs, please reach out via email or phone during regular support hours.